

Safety Data Sheet for Chemical Products

GB/T 16483-2008



VISIOMER® HPMA 98

Specification MPP0000000055

Version: 4.2/GHS

Revision Date: 3/12/2014

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1. IDENTIFICATION OF THE SUBSTANCE/PREPARATION AND OF THE COMPANY/UNDERTAKING

Trade Name: VISIOMER® HPMA 98
Chemical Name: hydroxypropyl methacrylate
Recommended Use(s): thermally crosslinkable paint resins, thermally crosslinkable emulsion polymers, textile fiber grating, reactive adhesives and reactive coatings, photopolymer plates and photoresists, reactive thinners and radiation curing, urethane (meth) acrylates.
Non-recommended use(s): preparations for nail extensions
Manufacturer: Evonik Specialty Chemicals (Shanghai) Co., Ltd.
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Effective on: 2014-03-12
National Emergency Call: 110, 119, 120

2. HAZARDS IDENTIFICATION

This substance is classified as hazardous acc. to GHS

GHS-Classification

Skin Sensitisation	Hazard category 1	H317
Serious eye damage/eye irritation	Hazard category 2 B	H320

GHS Label Elements

Signal word

Warning

Hazard symbols



Hazard statement

May cause an allergic skin reaction. (H317)
Causes eye irritation. (H320)

Precautionary statement

Safety notice (general)	Wear protective gloves/protective clothing/eye protection/face protection. (P280)
Precautionary Statement (Response)	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. (P305 + P351 + P338) IF ON SKIN: Wash with plenty of soap and water. (P302 + P352)
Precautionary Statement (Disposal)	Dispose of contents/container in accordance with local regulation. (P501)
Hazardous component(s) for labelling	hydroxypropyl methacrylate

3. COMPOSITION/INFORMATION ON INGREDIENTS

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Chemical nature: hydroxypropyl methacrylate

Hazardous Ingredients

Component	CAS-No.	Content	Hazard class / Hazard category / Hazard statement
hydroxypropyl methacrylate	27813-02-1	>= 98.0 %	Skin Sens. 1; H317 Eye Irrit. 2B; H319

4. FIRST AID MEASURES

General advice	Remove soiled, soaked clothing immediately. Medical treatment is necessary if symptoms occur which are obviously caused by skin or eye contact with the product or by inhalation of its vapours.
Inhalation	Move subject to fresh air and keep him calm. In case of complaints get medical treatment.
Skin contact	Wash off immediately with soap and water. If skin irritation occurs consult a physician.
Eye contact	Flush eyes thoroughly with a large amount of water and consult a physician.
Ingestion	Do not induce vomiting. Contact a doctor immediately.
Most important symptoms and effects, both acute and delayed	sensitization, Skin irritation, Eye irritation
Indication of any immediate medical attention and special treatment needed	No

5. FIRE-FIGHTING MEASURES

Suitable extinguishing media	foam, dry chemical, carbon dioxide
Extinguishing media which must not be used for safety reasons	water
Special hazards arising from the substance or mixture	May be released in case of fire: carbon monoxide, carbon dioxide, organic products of decomposition.
Advice for firefighters	Wear self-contained breathing apparatus.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions	Take care for adequate ventilation. Use personal protective clothing. Keep away sources of ignition. Use breathing apparatus if exposed to vapours/dust/mist/aerosol.
Environmental precautions	Prevent product from getting into drains/surface water/groundwater.
Methods for cleaning up	Larger quantities: Remove mechanically (by pumping). Use explosion-proof equipment! Smaller quantities and/or residues: Contain with absorbent material (e.g. sand, diatomaceous earth, acid absorbent, universal absorbent or sawdust). Dispose of in accordance with regulations.
Reference to other sections	For personal protection see section 8.

7. HANDLING AND STORAGE

Safe handling advice	Keep container tightly closed. Ensure the area is well ventilated.
Advice on protection against fire and explosion	In the event of fire, cool the endangered containers with water. When heated above the flash point and/or during spraying (atomizing), ignitable mixtures may form in air.
Requirements for storage areas and containers	Protect from light. Fill the container by approximately 90 % only as oxygen (air) is required for stabilisation. With large storage containers make sure the oxygen (air) supply is sufficient to ensure stability. Keep only in the original container at a temperature not exceeding 30 °C. Can polymerize with intense heat release.

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8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Protective measures	Do not inhale vapours. Avoid contact with eyes and skin.
Hygiene measures	Store work clothing separately. Remove soiled or soaked clothing immediately. Follow the usual good standards of occupational hygiene. Clean skin thoroughly after work; apply skin cream.
Respiratory protection	Breathing apparatus in case of high concentrations
Hand protection	butyl rubber gloves (0.7 mm), Break through time 480 min (EN 374) In practice, due to variable exposure conditions, this information can only be an aid to orientation for the selection of a suitable chemical protection glove. In particular, this information does not substitute suitability tests by the end user.
Splash protection	neoprene gloves
General information	Gloves should be replaced regularly, especially after extended contact with the product. For each work-place a suitable glove type has to be selected.
Eye protection	tightly fitting goggles
Skin and body protection	On handling of larger quantities: face mask, chemical-resistant boots and apron

9. PHYSICAL AND CHEMICAL PROPERTIES

Form	liquid
Colour	colourless
Odour	ester-like
Melting range	-90 °C (1013 hPa), OECD 102
Boiling Temperature	209 °C (1013 hPa), OECD 103
Flash point	101 °C (DIN 51758)
Ignition temperature	355 °C (DIN 51794)
Lower explosion limit	not available
Upper explosion limit	not available
Vapour pressure	0.11 hPa (20 °C) (OECD 104, dynamic method)
Density	1.03 g/cm ³ (20 °C)
Relative vapour density (related to air)	> 1 (20 °C)
Solubility in water	107 g/l (25 °C)
Solubility (qualitative)	miscible with most organic solvents
pH	6 (50 g/l) (20 °C)
n-Octanol/water partition coefficient	log Pow 0.97, (measured) source: literature
Viscosity (dynamic)	8.88 mm ² /s (20 °C) (DIN 51562)
Other information	none

10. STABILITY AND REACTIVITY

Reactivity	See "Chemical Stability"
Chemical Stability	No decomposition when used as directed.
Possibility of hazardous reactions	Polymerization with heat evolution may occur in the presence of radical forming substances (e.g. peroxides), reducing substances, and/or heavy metal ions.
Conditions to avoid	The product is normally supplied in a stabilized form. If the permissible storage period and/or storage temperature is exceeded, the product may polymerize with heat evolution.

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Incompatible materials Reactions with strong oxidizing agents.
Hazardous decomposition products None when used as directed.

11. TOXICOLOGICAL INFORMATION

toxicokinetics, metabolism and distribution	The substance is rapidly metabolized	
Acute Oral Toxicity	LD50 rat, OECD 401, limit test, no evidence for hazardous properties	> 2000 mg/kg
Acute Dermal Toxicity	LD50 rabbit:	> 5000 mg/kg
Caustic burning / irritation of skin	rabbit, 24 h, FDA 1959 Draize, occlusive, (own study)	not irritating - slightly irritating
Serious eye damage/eye irritation	rabbit, Draize, Eye irritant Category 2B (UN-GHS)	irritating
Respiratory/skin sensitization	guinea pig, In animal experiments the substance shows low resp. no ability as a sensitizer. There are indications of a sensitizing effect of the substance in man. Skin Sensitisation Category 1B (UN-GHS)	
Aspiration hazard	no evidence for hazardous properties (structure-activity-relationships) (analogy)	
Mutagenicity assessment	Positive as well as negative results in in vitro mutagenicity/ genotoxicity tests. No experimental indication of genotoxicity in vivo available. In summary not mutagenic according to internationally accepted criteria.	
Carcinogenicity	no specific test data available no evidence for hazardous properties (structure-activity-relationships) (analogy)	
Reprotoxicity / teratogenicity	No indications of toxic effects were observed in reproduction studies in animals. non-teratogenic, not embryotoxic	
Reprotoxicity / fertility	No indications of toxic effects were observed in reproduction studies in animals.	
Human health hazard assessment	CMR: no	
Specific Target Organ Toxicity - Single exposure	no evidence for hazardous properties	
Specific Target Organ Toxicity - Repeated exposure	no evidence for hazardous properties	
Toxicity on Repeated Administration	rat, oral, 49 d	NOAEL 300 mg/kg
General information	Avoid contact with the skin and eyes and inhalation of the product vapours.	

12. ECOLOGICAL INFORMATION

Aquatic toxicity, fish	LC50 Leuciscus idus, DIN 38412 section 15, 48 h (own study)	493 mg/l
Aquatic toxicity, invertebrates	EC50 Daphnia magna, OECD 202, 48 h NOEC Daphnia magna, OECD 202, 21 d	380 mg/l 24.1 mg/l
Aquatic toxicity, algae / aquatic plants	EC50 Pseudokirchneriella subcapitata, OECD 201, 72 h NOEC Pseudokirchneriella subcapitata, OECD 201, 72 h	> 97.2 mg/l 97.2 mg/l

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Toxicity in microorganisms	EC10 Pseudomonas putida, Bringmann-Kühn , 16 h (own study)	1140 mg/l
Persistence and degradability	Readily biodegradable, according to appropriate OECD test. The substance photodegrades rapidly when exposed to air.	
Biodegradability	readily degradable, 79/831/EEC, Modified OECD Screening Test, 28 d (own study)	ca. 94 %

Bioaccumulation

Bioaccumulation	Accumulation in organisms is not expected due to the coefficient of distribution of n-octanol in water (log Pow).	
Mobility	Very sparingly volatile from the aqueous phase. Binding to the solid soil phase, sediment or clarification sludge is not expected. The substance is distributed mainly into the water phase and the soil.	
PBT and vPvB assessment	PBT: no vPvB: no	
General Information	Prevent substance from entering soil, natural bodies of water and sewer systems.	

13. DISPOSAL CONSIDERATIONS

Product	Waste is hazardous. It must be disposed of in accordance with the regulations after consultation of the competent local authorities and the disposal company in a suitable and licensed facility.
Uncleaned packaging	Contaminated packaging should be emptied optimally and after appropriate professional cleansing may be taken for reuse. Packaging that cannot be cleaned should be disposed of professionally. Uncontaminated packaging may be taken for recycling.

14. TRANSPORT INFORMATION

Not dangerous according to transport regulations.

15. REGULATORY INFORMATION

Legislations of Safety Administration of Chemicals

National Legislation	Status
MEP: Inventory of the Existing Chemical Substances in China:	listed/registered

16. OTHER INFORMATION

Other information	The product is normally supplied in a stabilized form. If the permissible storage period and/or storage temperature is exceeded, the product may polymerize with heat evolution.
Relevant H phrases from chapter 3	hydroxypropyl methacrylate H317- May cause an allergic skin reaction. H320- Causes eye irritation.
Filling Time	See header for the version and status of this MSDS
Filling Department	Evonik Industries AG
Regulation Compliance Check:	Environmental, Safety, Health, and Quality Department of Evonik China is responsible for the product's regulation compliance check in China.

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Version and Issue Date

Version 4.0 2013-09-20

Revision Statement

Latest revisions are marked in spaces; this version is the latest.

References

- [1]MEP: Inventory of Existing Chemical Substance in China (IECSC)
- [2]SAWS: Hyper Toxic Chemical Inventory
- [3]SAWS: List of Hazardous Chemical
- [4]SAWS: Differentiation of Great Danger Source
- [5]MEP: National Hazardous Wastes Inventory
- [6]MOH: List of Highly Toxic Substance
- [7]State department: Regulation on Management of Precursor of Narcotic Chemicals
- [8]MOH: Occupational Exposure Limits for Hazardous Agents in the Workplace
- [9]List of Dangerous Goods
- [10]MEP, Customs: Catalogue of Toxic Chemicals Prohibited or Strictly Controlled to Import or Export

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